

Review on geometric optics and interference

Phys2500J

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Overview

1. Reflection and Refraction

2. Dispersion

3. Polarization

4. Interference

Question?

Why the image in the mirror is opposite in the left and right instead of up and down?

Reflection and Refraction

- Reflection
 - Specular reflection
 - Diffuse reflection
 - Laws of reflection $\theta_i = \theta_r$
- Refraction
 - Snell's Law: $n_i \sin(\theta_i) = n_t \sin(\theta_t)$
 - Relation between index and the wavelength
- Huygens's principle
- Total internal reflection
 - Only happens if $n_a > n_b$
 - Critical Angle $\theta_{critical} = \arcsin(n_b/n_a)$
 - Application: Optical Fibers, Pentaprism, Sparkle of diamonds

More thoughts on refraction

Dispersion



Figure: Dark side of the moon–Pink Floyd

Dispersion

Definition

The index of refraction depends on the wavelength of the light

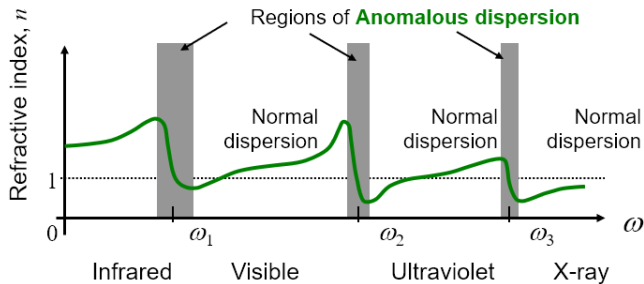


Figure: Refractive Index vs. Frequency

Examples for dispersion

Examples

Rainbows

Point out the different between Primary rainbow and Secondary rainbow?

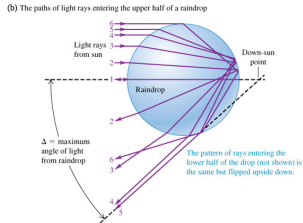


Figure: Path of light rays entering the upperhalf of a raindrop

Polarization

Definition

An electromagnetic wave is linearly polarized if the electric field has only one component.

- Polarizing filter, polarizer
- Malus's Law
- Brewster's Angle
- Application: 3D glasses, sunglasses
- Birefringence: halfwave plate, quarter-wave plate

Brewster's Angle

Scattering

Why clouds are white? (seems like questions from One Hundred Thousand Whys)

Reasons: Clouds are white because they scatter all wavelengths efficiently

Overview

Key concepts of interference

- Wave function. (Recommend written in Euler function form)
- Superposition principle
- Phase difference & Optical Pathlength difference (OPD)

Two-source interference

Two-source interference intensity

Result

- Constructive interference: $\delta = 2m\pi$
- Destructive interference: $\delta = (2m + 1)\pi$

Intensity

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\delta)$$

Fringe visibility

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2} \leq 1$$

Thin film interference

π phase shift

Interference model

- Wavefront Splitting
 - Young's Double-slit interferometer
 - Fresnel's biprism
 - Lloyd's mirror
- Amplitude Splitting
 - Dielectric Film: Double-Beam Interference
 - Fringe of Equal Inclination: Haidinger Fringes
 - Fringe of Equal Thickness: Newton Rings
 - Michelson Interferometer

Problem Set

The End